Multivariate Analysis

Canonical Correlation Analysis

Johnson, RA & Wichern, DW. "Applied Multivariate Analysis", Chapter 10, 6th edition, Pearson International Edition, 2007.

Canonical Correlation analysis aims to describe relationships between two sets of variables, $\boldsymbol{X}^{(1)}_{p\times 1}$and $\boldsymbol{X}^{(2)}_{q\times 1}$.

Examples include the association between indicators of arithmetic speed and indicators of arithmetic power; or university performance and pre-university achievement. It does so by focusing on the correlation between LINEAR combinations of variables in one set to LINEAR combinations of variables in another set. Hence in general the analysis involves

Determine the pair of linear combinations having the largest correlation,

$$U_1 = a_{11}X_1^{(1)} + a_{12}X_2^{(1)} + \cdots + a_{1p}X_p^{(1)}$$

and

$$V_1 = b_{11}X_1^{(2)} + b_{12}X_2^{(2)} + \cdots + b_{1q}X_q^{(2)}.$$

Determine the pair of linear combinations having the largest correlation among all pairs uncorrelated with the initial pair.

And so on.

The pairs of linear combinations are called CANONICAL VARIABLES and their correlations are called CANONICAL CORRELATIONS, which measure the strength of association between the two sets of variables.

The overall aim is to attempt to concentrate a high-dimensional relationship between two sets of variables into a few pairs of canonical variables.

Objective in terms of maximizing correlation matrix

Given the covariance matrix between the variables within a group and between variables from different groups,

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11(p\times p)} & \vdots & \boldsymbol{\Sigma}_{12(p\times q)} \\ \cdots & \vdots & \cdots \\ \boldsymbol{\Sigma}_{21(q\times p)} & \vdots & \Sigma_{22(q\times q)} \end{bmatrix}$$

Note the partitioning of the covariance matrix in terms of the 2 subsets of variables.

Let the linear combinations be $\boldsymbol{U} = \boldsymbol{a}'\boldsymbol{X}^{(\mathbf{1})}$ and $\boldsymbol{V} = \boldsymbol{b}'\boldsymbol{X}^{(\mathbf{2})}$ then

$$Var(\boldsymbol{U}) = \boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{a}$$
$$Var(\boldsymbol{V}) = \boldsymbol{b}'\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{b}$$
$$Covar(\boldsymbol{U},\boldsymbol{V}) = \boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{b}$$

These expressions just follow the normal rules for finding the variance of a random variable pre-multiplied by a constant, e.g. Var(aX)=a^2Var(X), etc.

We wish to find vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ such that

$$Corr(\boldsymbol{U},\boldsymbol{V}) = \frac{\boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{b}}{\sqrt{\boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{a}}\sqrt{\boldsymbol{b}'\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{b}}}$$

And this is of course how we calculate the correlation as the ratio of the covariance divided by the product of the 2 sd’s.

is as large as possible.

Solution to maximization problem

It can be shown that the $\boldsymbol{a}$ and **b** that will maximize the $Corr(U,V) = \rho_1^*$ are given by

$$U_1 = \underbrace{\boldsymbol{e}_1'\boldsymbol{\Sigma}_{11}^{-1/2}}_{\boldsymbol{a}_1'}\boldsymbol{X}^{(1)} \text{ and } V_1 = \underbrace{\boldsymbol{f}_1'\boldsymbol{\Sigma}_{22}^{-1/2}}_{\boldsymbol{b}_1'}\boldsymbol{X}^{(2)}$$

and for the k^{th} pair,

$$U_k = \underbrace{\boldsymbol{e}_k'\boldsymbol{\Sigma}_{11}^{-1/2}}_{\boldsymbol{a}_k'}\boldsymbol{X}^{(1)} \text{ and } V_k = \underbrace{\boldsymbol{f}_k'\boldsymbol{\Sigma}_{22}^{-1/2}}_{\boldsymbol{b}_k'}\boldsymbol{X}^{(2)}$$

So similar to PCA but scaled by $\Sigma_{11}^{-1/2}$ or $\Sigma_{22}^{-1/2}$

where $\rho_1^{*2} \geq \rho_2^{*2} \geq \cdots \geq \rho_p^{*2}$ are the eigenvalues of

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$$

We will unpack this product a little more later.

and $\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_p$ are the associated eigenvectors.

Equivalently, $\rho_1^{*2} \geq \rho_2^{*2} \geq \cdots \geq \rho_p^{*2}$ are the eigenvalues of

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$$

So you only need to calculate the eigendecomposition of one product and then you can obtain the f_i from the e_i. You will still need to divide by variance to ensure unit variance.

and associated eigenvalues $\boldsymbol{f}_1, \boldsymbol{f}_2, \dots, \boldsymbol{f}_p$ and each $\boldsymbol{f}_i$ is proportional to $\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{e}_i$. Needs standardization so as to have unit variance – see r-code later on.

Properties of Canonical Variates

The canonical covariates have the following properties:

$$Var(U_k) = Var(V_k) = 1$$
$$Cov(U_k U_l) = Corr(U_k U_l) = 0$$
$$Cov(V_k V_l) = Corr(V_k V_l) = 0$$
$$Cov(U_k V_l) = Corr(U_k V_l) = 0$$

for $k \neq l$.

Standardized unit variances and independence among U and V of different linear combinations when k ne l.

If the original variables are standardized, the canonical variates are of the form

$$U_k = \boldsymbol{a}'_k \boldsymbol{Z}^{(1)} = \boldsymbol{e}'_k \boldsymbol{\rho}_{11}^{-1/2} \boldsymbol{Z}^{(1)} \text{ and } V_k = \boldsymbol{b}'_k \boldsymbol{Z}^{(2)} = \boldsymbol{f}'_k \boldsymbol{\rho}_{22}^{-1/2} \boldsymbol{Z}^{(2)}$$

Here Z represent the standardized variables and hence the covariance matrix of the Z is actually the correlation matrix of the X variables.

where

$\boldsymbol{\rho}_{11}$ and $\boldsymbol{\rho}_{22}$ are sub-matrices of the correlation matrix.

Interpretation

The canonical loadings can be thought of as the correlation between a variable set and its corresponding canonical variate and can thus be used to interpret the canonical variates, even though there is no information in these correlations about how one variable list contributes jointly to the canonical correlation with the other, i.e., they provide only univariate information.

The loadings can also be interpreted like regression coefficients, measuring the impact on the canonical variate of a one unit increase in the specific measured variable.

One way of thinking about Canonical Correlation Analysis is that every bit of correlation is wrung out of the original $X^{(1)}$ and $X^{(2)}$ variables and deposited in an orderly fashion into pairs of new variables U_k and V_k which have a special correlation structure.

Can also compute actual correlations between canonical variates and the original variables using $\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{X}^{(1)}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{V}_{\mathbf{11}}^{-\mathbf{1/2}}$ where $\boldsymbol{V}_{\mathbf{11}} = diag\left(Var\left(X_k^{(1)}\right)\right) = diag(\sigma_{kk})$ or $\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{Z}^{(1)}} = \boldsymbol{A}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{11}}$ where $\boldsymbol{A}_{\boldsymbol{z}}$refers to canonical variates based on correlation matrix. Similarly,

$$\boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{X}^{(2)}} = \boldsymbol{B}\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{V}_{\mathbf{22}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{Z}^{(2)}} = \boldsymbol{B}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{22}}$$
$$\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{X}^{(2)}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{V}_{\mathbf{22}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{Z}^{(2)}} = \boldsymbol{A}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{12}}$$
$$\boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{X}^{(1)}} = \boldsymbol{B}\boldsymbol{\Sigma}_{\mathbf{21}}\boldsymbol{V}_{\mathbf{11}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{Z}^{(1)}} = \boldsymbol{B}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{21}}$$

where $\boldsymbol{V}$ and $\boldsymbol{\rho}$ are the covariance and correlation matrices of $\boldsymbol{X}$.

Interpretation of R

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$$

We are taking eigendecomposition of matrix of squared correlation coefficients as opposed to of covariance matrix which was the case for PCA.

When $p = 1$, i.e., just one $X^{(1)} = Y$,

R reduces to the squared correlation coefficient of $X^{(1)} = Y$ with the best linear predictor of $X^{(1)}$ using $X_1^{(2)}, X_2^{(2)}, \dots, X_q^{(2)}$

$$R = \rho^2_{Y.X_1^{(2)} \dots X_q^{(2)}} = \frac{\sigma_{YX}^T \Sigma_{XX}^{-1} \sigma_{XY}}{\sigma_Y^2}$$

The simplified form of R when we just have one variable in the first group.

And when both $p = 1$ and $q = 1$,

$R = \rho^2 = \frac{\sigma_{XY}^2}{\sigma_X^2 \sigma_Y^2}$, the squared correlation coefficient between X and Y.

Thus the j^{th} canonical correlation coefficient, ρ_j , can be interpreted as either the multiple correlation coefficient of V_j (the linear combination of $X^{(2)}$) with $X^{(1)}$ or of U_j (the linear combination of $X^{(1)}$) with $X^{(2)}$.

So ρ_j can be viewed as either

the proportion of the variance of U_j attributable to its linear regression on $X^{(2)}$

or of V_j attributable to its linear regression on $X^{(1)}$.

Sample Canonical Variates and Correlations

When we have a sample of n observations on each of the $p + q$ variables in $X^{(1)}$ and $X^{(2)}$ we work with the sample covariance matrix

$$\boldsymbol{S} = \begin{bmatrix} \boldsymbol{S}_{11(p\times p)} & \vdots & \boldsymbol{S}_{12(p\times q)} \\ \cdots & \vdots & \cdots \\ \boldsymbol{S}_{21(q\times p)} & \vdots & \boldsymbol{S}_{22(q\times q)} \end{bmatrix}$$

or the corresponding correlation matrix R.

Interpreting a Canonical Correlation Analysis

The canonical variates are artificial, they have no physical meaning.

The units of the canonical coefficients are proportional to the units of the variables in the two variable sets.

If the original variables are standardized (or, equivalently, we use the correlation matrices), the canonical coefficients have no units of measurement.

The canonical variates can be "identified" by looking at the correlations between the canonical variates and the original variables, (similar to a factor analysis).

Class Exercise 1

Use R and program from first principles to calculate the first pair of canonical variates of the covariance matrix and the canonical correlation.

```
> Cov
      [,1] [,2] [,3] [,4]
[1,]  100 0.00 0.00    0
[2,]    0 1.00 0.95    0
[3,]    0 0.95 1.00    0
[4,]    0 0.00 0.00  100
```

Save annotated r-code and output in a pdf file labelled studentno_CCA1.pdf and submit via VULA.

Class Exercise 2

Return to the Stock Price data and perform a canonical correlation analysis with $X^{(1)}$ being the banking rates of returns and $X^{(2)}$ being the rates of returns for the oil companies.

Save annotated r-code and output in a pdf file labelled studentno_CCA2.pdf and submit via VULA.

To Summarize (What you should have learnt):

- Definition of Canonical Correlation Analysis (CCA), Canonical variates (CCVs) and Canonical Correlation (Slide1)
- Partitioning of the covariance matrix and calculation of correlation between 2 linear combinations (Slide 2)
- Expression for canonical coefficients and introduction of R and relationship between e and f (Slide 3)
- Properties of U and V and formulation in terms of correlation matrix (standardized variables) (Slide 4)
- Guide to interpretation and distinction between canonical coefficients and correlations between measured variables and canonical variables (Slide 5)
- More about R (Slide 6)
- R-code for CCA from first principles (Slides 7 to 10)
- Example using R function cc from R library CCA (Slides 13 to 18)